

Unit 1 Vocabulary (Alphabetical Order)

Abstraction - One of the key components of computational thinking. Abstraction removes unnecessary details, focusing on just the essential elements of an idea. For example, an email address such as dave@gmail.com is an abstraction; gmail.com is not a computer; it's a collection of many computers distributed around the world. But you don't need to know those details to send a message to Dave; you just need to know his email address.

AI (Artificial Intelligence) - a branch of computer science concerned with techniques for getting computers to do things that normally require human intelligence to do.

Autonomous - Self-determined, aware of the world, self-governed, ability to set goals and choose actions to achieve those goals.

Graph - In computer science a graph is a collection of nodes connected by links. The nodes are typically drawn as circles and the links are drawn as lines connecting one circle to another. Graphs can be used to represent many different types of things. For example, a road map can be represented as a graph where the nodes are cities and the links are highways that connect one city to another.

Hardware - physical devices such as computers, phones, or tablets, or physical components such as computer chips, disk drives, or chargers. Compare with *software*.

Object recognition - A problem in computer vision: recognizing objects such as people, animals, cars, and toasters. This problem is hard because not only do people and cars come in many shapes and colors, but even a single car looks quite different when viewed from different angles.

LIDAR - Light Direction and Ranging. Similar to radar, LIDAR detects objects by bouncing a spot of laser light off them and measuring the time it takes the light to be reflected back. LIDAR cannot see color.

Perception - Extraction of meaning from sensory signals. An old-fashioned camera records images by sensing light, but it does not "see". (Some digital cameras with built-in face detection do "see" to a limited extent.)

Radar - Radio Direction and Ranging. A technique for detecting objects by bouncing radio signals off them and measuring the time it takes for the signal to return. This gives the distance to the object. Knowing where the antenna was pointed when the signal was sent gives the angle to the object. Radar was originally used to detect aircraft, but today is also used by self-driving cars to detect obstacles. Radio waves aren't affected by fog,

so radar can “see” in low visibility conditions where regular cameras cannot.

Representation - information from the real world for a computer to understand and then utilize this knowledge to solve complex real-life problems

Research - Systematic investigation and study. In this course, students can research a topic by looking for information on the Internet. Another way to do research is by running experiments.

Robot - a type of automated machine that can execute specific tasks with little or no human intervention

Sensing - Measuring physical phenomena such as light, sound, temperature, or pressure.

Sensor - A device that detects or measures a physical phenomenon

SONAR - Sonic Navigation and Ranging. Sonar works similar to radar except it uses sound waves instead of radio waves. By sending out a pulse of high-frequency sound and measuring the time it takes for the sound to be reflected back, sonar can measure distance to an object. Sonar is used by submarines to detect other vessels under water, but it also works in air. Self-driving cars may use sonar to measure distance to other cars when parallel parking. Dolphins and bats also use sonar.

Software - The instructions that make computing devices do useful things. Software includes operating systems such as Windows or MacOS, and applications such as web browsers, video players, and computer games. Compare with *hardware*.

Subsystems - components of a complex device, e.g., a self-driving car would have subsystems for navigation, collision avoidance, steering control, etc.

Trees - In computer science, a tree is a structure composed of nodes and links. Trees are a special type of *graph*, where each node has exactly one parent node except for the “root” node at the top of the tree, which has no parent. Trees are used to represent many types of **A stems or machines that mimic human intelligence to perform tasks and can iteratively improve themselves based on the information they collect** things. Search trees are constructed by search algorithms (such as breadth-first search) to solve certain kinds of reasoning problems.